

STATE OF COLORADO

DIVISION OF MINERALS AND GEOLOGY

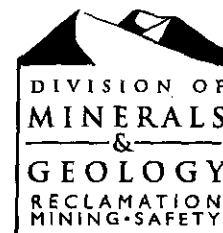
Department of Natural Resources

1313 Sherman St., Room 215

Denver, Colorado 80203

Phone: (303) 866-3567

FAX: (303) 832-8106



November 7, 2002

Mr. Steve Mossberg
CEMEX
P.O. Box 529
Lyons, CO 80540

Bill Owens
Governor

Greg E. Walcher
Executive Director

Ronald W. Cattany
Acting Division Director

RE: CEMEX, Inc., Lyons Mine, Permit No. M-1977-208
Groundwater Monitoring for CKD

Dear Mr. Mossberg,

The Division of Minerals and Geology has reviewed the groundwater monitoring requirements for the CKD as specified in Technical Revision 01 (TR-01). Attached is a memo from Harry Posey of DMG providing a synopsis and recommendation for CKD groundwater monitoring.

In summary, the numeric protection levels (NPL's) are as follows:

pH	6.5-8.5 std. units
Conductivity	Measure and report
Se	0.02 mg/L
TI	0.002 mg/L

Monitoring must be reported twice yearly, and submitted with the Annual Reclamation Report. The Division recommends that more frequent monitoring be conducted for pH and conductivity than what was approved in TR-01. In the event pH or conductivity increases, the Division should be notified and more frequent monitoring for possible metal contaminants should be conducted and reported.

In regard to the test burn of tires in the plant, CEMEX will need to subject the ash to representative leach tests. The particular leach test can be determined by the operator with DMG approval of the method. The SPLP EPA Method 1312, analyzed for soluble organics, metals, SO₄, and pH might be an acceptable method.

If you have any questions, please do not hesitate to contact me at 303-866-4943.

Sincerely,

Erica S. Crosby
Environmental Protection Specialist

Enclosure (Memo from Harry Posey; DMG)

cc: Carl Mount; DMG
Harry Posey; DMG
John Lohr; CEMEX (w/enclosure)

STATE OF COLORADO

DIVISION OF MINERALS AND GEOLOGY

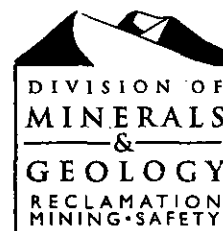
Department of Natural Resources

1313 Sherman St., Room 215

Denver, Colorado 80203

Phone: (303) 866-3567

FAX: (303) 832-8106



November 6, 2002

To: Erica Crosby

From: Harry Posey 

Subject: CKD monitoring; Cemex, Lyons Quarry, M-1977-208, TR-01

Bill Owens
Governor

Greg E. Walcher
Executive Director

Ronald W. Cattany
Acting Division Director

Reviewing the file, it appears that DMG set out to establish baseline conditions for CKD weathering at the captioned operation. The following sequence, from 1999 through 2001, synthesizes the monitoring plan and the results, and *proposes a set of numeric protection levels (NPL's).*

Synopsis:

1. DMG required leach testing of CKD. Results indicated elevated selenium (Se), thallium (Tl) and high pH. Conductivity, which is not regulated, was selected as an indicator parameter.
2. DMG selected these parameters for monitoring plus conductivity as an indicator for monitoring.
3. DMG also required that CKD be sprayed with water to promote crystallization.
4. The company was to submit 5 quarters of baseline analyses from A-Pit pond for Se, Tl, pH, and conductivity. This was accomplished.
5. ~~DMG was to consider the 5 quarters of baseline monitoring and establish NPL's. That proposal is forwarded in this memo.~~
6. Upon establishment of NPL's, the Company was approved to reduce monitoring to twice yearly.
7. Based on monitoring results, Se and Tl consistently reported below regulated values. Data were good, detection levels were proper.
8. In all cases, pH was within the allowable range for groundwater.
9. Conductivity was consistently high. Conductivity is not regulated.

Ordinarily, DMG requires 8 samples over the 5 quarters of baseline in order to apply Dixon statistical methods to determine NPL's. Only 5 samples were analyzed. However, as the metal values fell below the regulated values and as pH was within the allowable range for groundwater uses, I would suggest that we do not need 8 samples. Following are the results:

Reporting Date	Values			
	pH (s.u.)	Cond (uS/cm)	Se (mg/L)	TI (mg/L)
5/5/00	7.93	2520	0.004	BDL
7/17/00	8.00	2520	0.005	BDL
10/12/00	8.00	2470	BDL	BDL
1/28/01	8.03	2720	BDL	BDL
4/27/01	8.00	2630	BDL	BDL
Table value standard – drinking water	6.5-8.5	n/a	0.05	0.002
Table value standard – agriculture	6.5-8.5	n/a	0.02	n/a

Based on the above, I would recommend that NPL's be established at the most restrictive water quality standards. In this case, the most stringent use standards would be agriculture.

Proposed NPL's	
pH	6.5- 8.5 std. units
Conductivity	Measure and report
Se	0.02 mg/L
TI	0.002 mg/L

As provided previously, monitoring should be reported twice yearly.

It is recommended that more frequent monitoring be conducted by the company for pH and conductivity. This would have to be voluntary as the Division has approved twice yearly sampling. In the event pH or conductivity increases, the division should be notified and more frequent monitoring for possible metal contaminants should be conducted and reported.

Cc: Bruce Humphries
Carl Mount